Displacement-Time Graphs

Exam code: 9MA0

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Displacement-time graphs

What is a displacement-time graph?

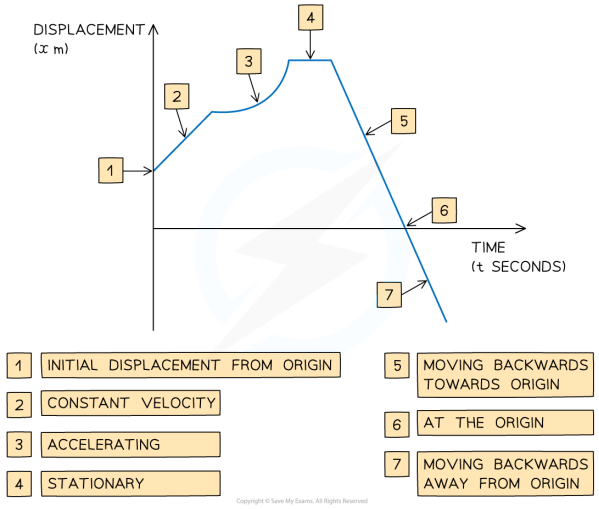
- **Displacement-time** graphs show the displacement of an object from a **fixed origin** as it moves in a **straight line**
- They show **displacement** (on the vertical axis) against **time** (on the horizontal axis)
- Displacement-time graphs can go **below the horizontal axis** whereas **distance-time graphs** can not
  - Distance can **not** be negative whereas displacement can be

What are the key features of a displacement-time graph?

- The **gradient** of the graph equals the velocity of the object
  - A **positive gradient** means the object is travelling **forwards**
  - A **negative gradient** means that the object is travelling **backwards**
  - The **steeper** the line, the **greater** the **speed**
- A **straight line** shows that the object is moving at a **constant velocity**
- A **curved line** shows that the object is **accelerating** or **decelerating**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



How do I find the average speed and the average velocity of a journey?

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Time taken}}$$

- The average speed can **not** be **negative** as speed is a **scalar** quantity and can only take a positive value

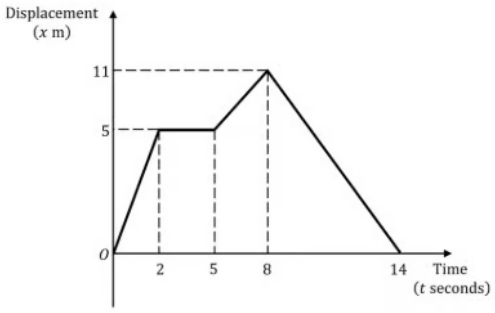
$$\text{Average velocity} = \frac{\text{Displacement from starting point}}{\text{Time taken}}$$

- The average velocity is a **vector** quantity and can be **positive**, **zero** or **negative**

Worked Example

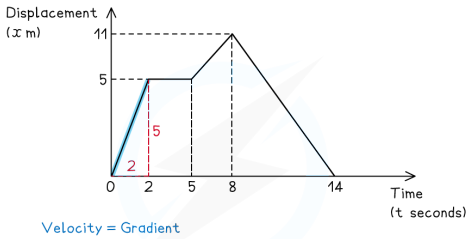
An athlete is training by jogging in a straight line.

The displacement-time graph below shows the displacement of the athlete from her starting point throughout her motion.



Answer:

a)



Velocity = Gradient

Velocity = $\frac{5}{2}$

Velocity = 2.5 ms^{-1}

Remember units!

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(b) Describe the motion of the athlete between the times of 2 seconds and 5 seconds.

Answer:

b)

Horizontal line means stationary

The athlete is stationary

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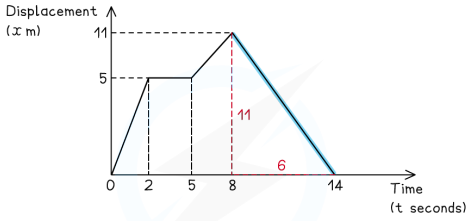
(c) Calculate the velocity of the athlete at 10 seconds.

Answer:

c)



Velocity = Gradient

Velocity = $\frac{0 - 11}{14 - 8}$

Velocity = $-\frac{11}{6}$ ms^{-1} = -1.83 ms^{-1} (3 sf)

Velocity is negative as the gradient is negative

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(d) Find the total distance travelled by the athlete during the 14 seconds.

Answer:

Examiner Tips and Tricks

- Be careful to spot if you are working with a **displacement-time graph** or a **velocity-time graph**.
- Be careful to spot if you are working with a displacement-time graph or a distance-time graph

walk starting 100 m from their house.

- Distance is a scalar so it can **not** be negative whereas displacement from a starting point can be.
- Think about the units when calculating, make sure they are consistent.

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